

Computer Forensics Case Studies

Case Study 1 Investigating Harassment Email Traffic With Wireshark

Student Assessment Brief

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SBTDF204_CaseStudy1_2025/FWSD/11478pdf

Assessment detail	Information
Course	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 1 Individual forensic investigation
Case title	Investigating Harassment Email Traffic With Wireshark
Submission format	Written report with supporting screenshots and evidence log
Deadline and maximum score	As published in E-Campus

This individual case study asks you to investigate a simulated harassment-email complaint using supplied email material, a class roster, and a packet capture. Your report must show how you preserved the evidence, found the relevant web activity in Wireshark, and assessed whether the traffic supports attribution to a student in Chemistry 109. Do not assume that an IP address proves identity. Your conclusion must follow the evidence and explain its limits.

Case scenario

Lily Tuckrige, a Chemistry Department teacher, reports repeated harassing emails sent to her personal Yahoo Mail account. Full headers from an earlier message point to an IP address in a shared student residence where an open wireless router is in use. On Monday 21 July, a later message arrived through the web service willselfdestruct.com. The service briefly displayed the message and then reported that it had been destroyed. The university captured traffic at the residence network boundary and asks you to determine whether the available evidence supports attribution to a student in Chemistry 109.

Case materials

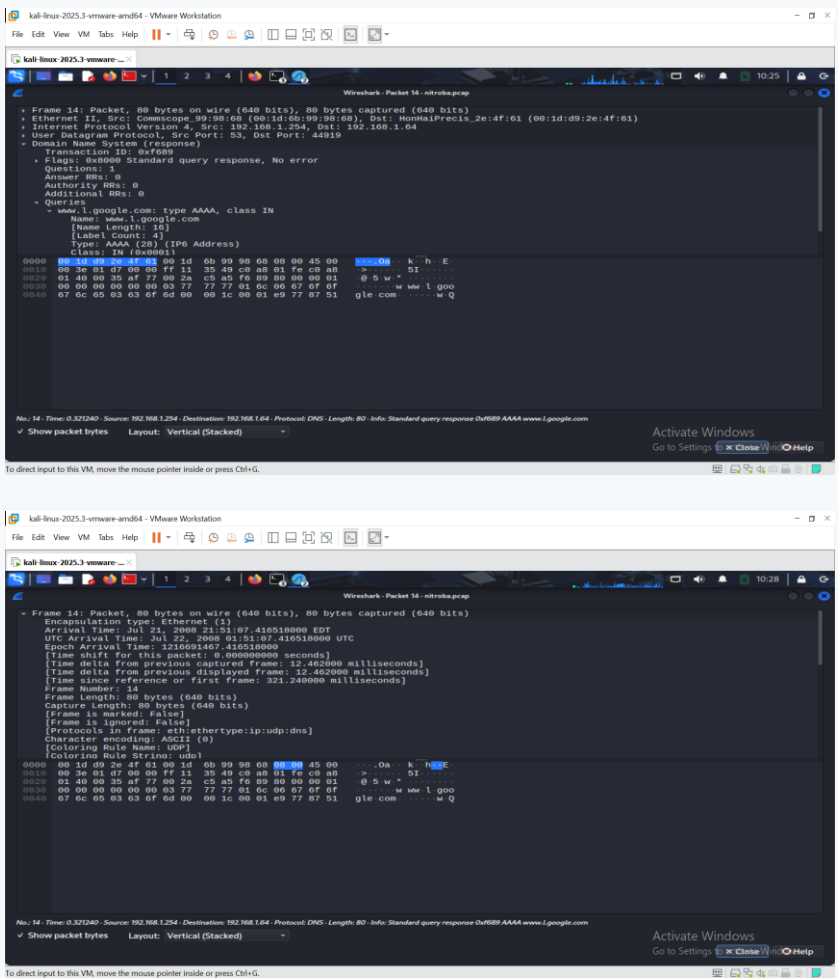
Use the scenario-provided first email, full headers, second email, class roster, and network traffic capture.

Scenario page: digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario

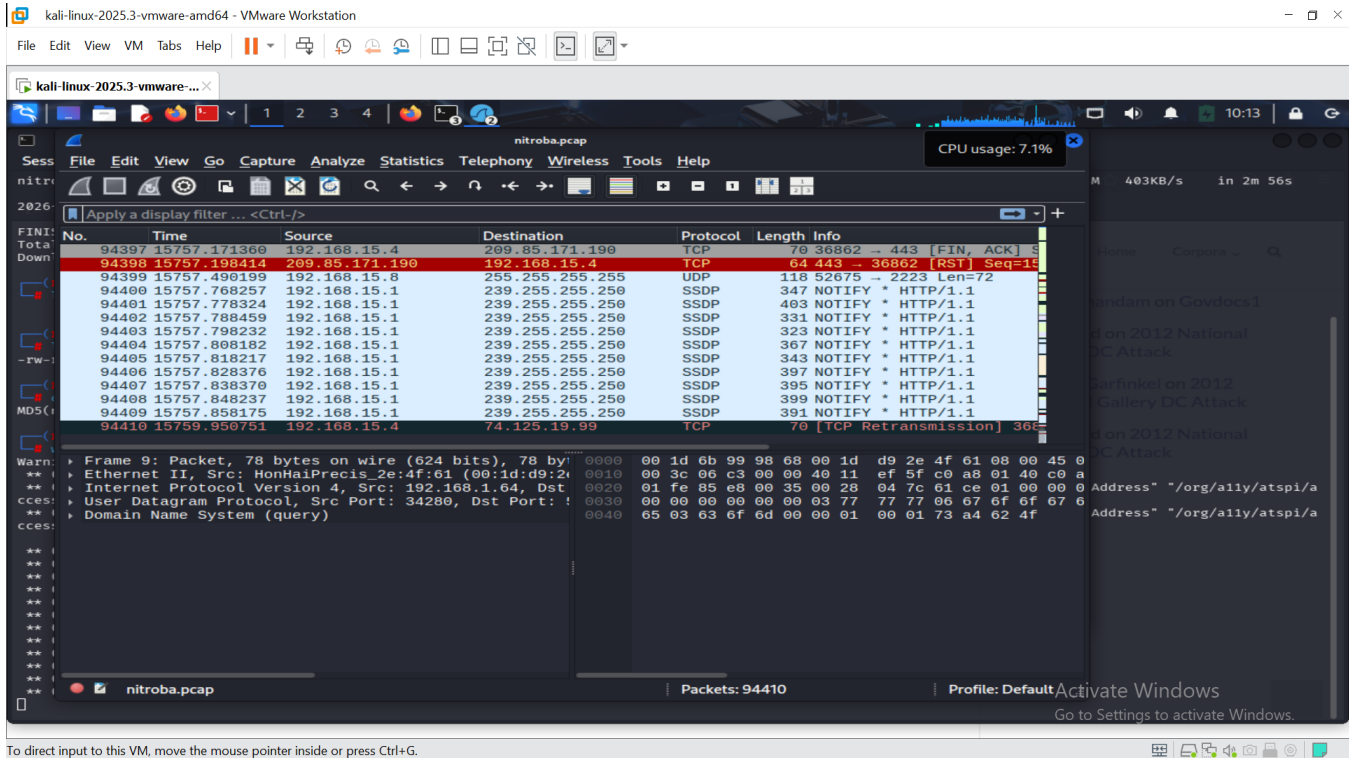
Traffic capture: digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap

If the source page also provides a published checksum, compare it with the hash you calculate. If it does not, record your calculated SHA-256 value in the report so another examiner can identify the evidence file you analysed.

What you must submit

Item	Required output	What to provide
1	Forensic working copy	Download the scenario traffic capture and work only from your copied evidence file. Record the original file name, source link, file size, download date, and SHA-256 value. SHA256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
2	Wireshark analysis	Use display filters, packet inspection, and Follow TCP Stream where appropriate. Preserve packet numbers and exact timestamps for every material finding. Packet number: 80 bytes on wire and (640 bits) Exact timestamp: 
3	Evidence-based conclusion	State only what the traffic supports. Separate direct observations, reasonable inferences, and limitations caused by the shared open wireless network.

Item	Required output	What to provide
4	Student report	Submit one report in PDF or DOCX format, with an evidence log and clearly labelled screenshots in an appendix or separate ZIP file.

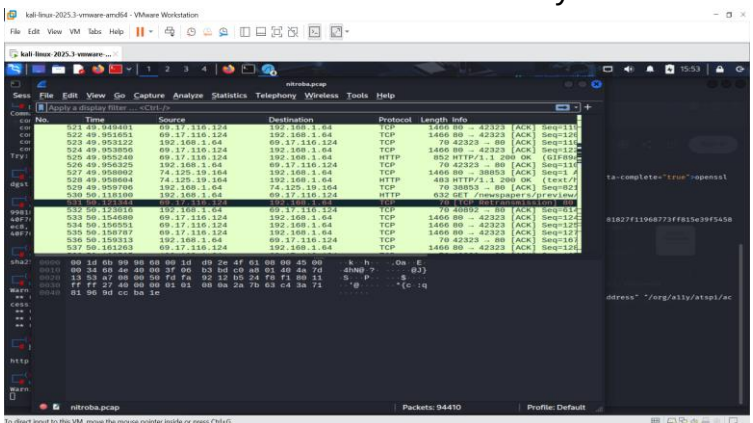
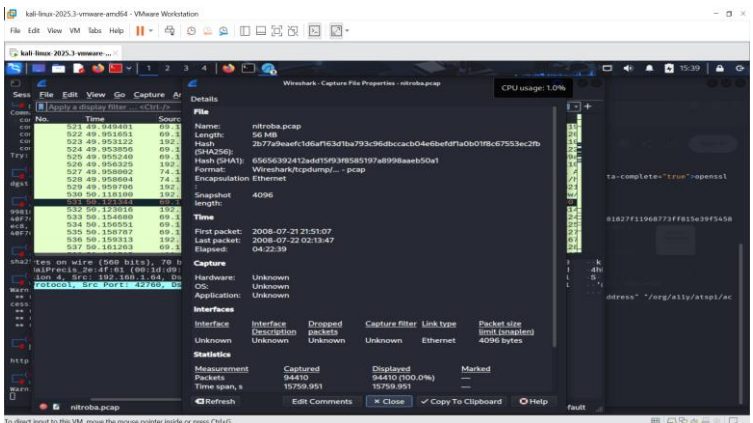


Investigation task

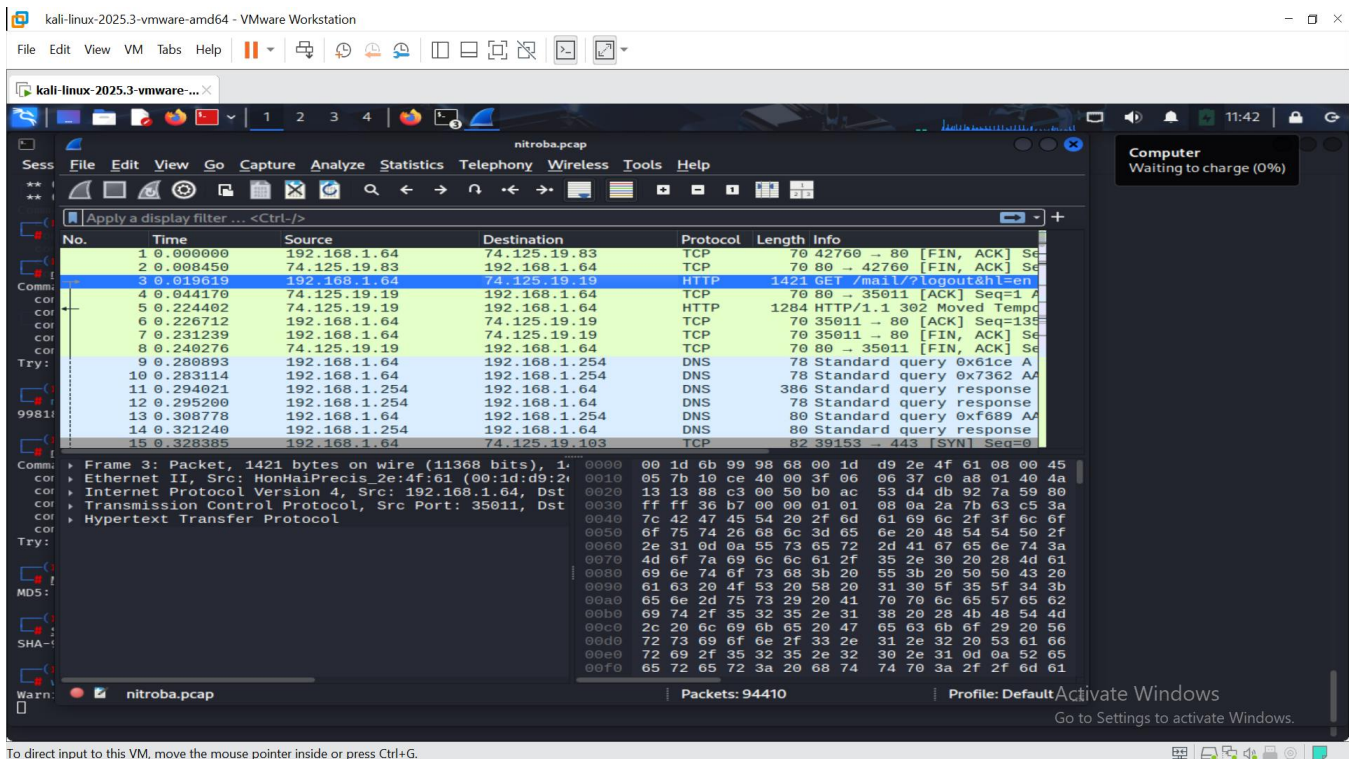
Conduct a methodical forensic review of the packet capture. Use the questions below as the structure for your findings. For every material finding, provide the Wireshark display filter, packet number or stream reference, timestamp, and a short explanation of why the evidence matters. You may use the filters as starting points, but test and adapt them against the actual capture.

No	Investigation question	Minimum evidence to discuss
1	What file did you acquire, and how did you preserve its integrity?	Give the file name: nitroba.pcap Source: Size: 54M SHA-256 value2b77a9eaefc1d6af163d1ba793c96dbccacb 04e6befdf1a0b01f8c67553ec2fb The method used to create a working file is kali command line

No	Investigation question	Minimum evidence to discuss
2	Which client system contacted the web service?	Identify the client IP: 192.168.1.64 the service IP: 74.125.19.83 Cite the packet number:94410 Filter that support each value are the Source ip 192.168.1.64 and The Destination: 4.125.19.83
3	What evidence links the client to the harassment message?	Show the relevant HTTP POST and explain how its content, form field, URI, or stream correlates with the reported message.
4	Which device made the relevant request?	Identify the MAC address associated with the client traffic: 00:1d:d9:2e:4f:61 and explain why a shared IP address alone cannot prove identity. Answer: A shared IP address cannot uniquely identify the person because - Multiple users may share a NAT gateway and DHCP may reassign addresses over time. Also, Multiple devices can use the same external public IP. - VPNs and proxies can mask actual endpoints. - P evidence identifies a device connection, not a specific individual. On that note, the MAC address helps identify the local network interface that generated the traffic, and identifies a device, not necessarily the person using it.

No	Investigation question	Minimum evidence to discuss
5	Who is associated with the device?	Use only scenario-provided cookie or browser evidence to link the device to an identity.  State the packet evidence: 94410 and avoid unsupported assumptions.
6	Is that person on the Chem 109 roster?	Compare the supported identity with the provided roster and show the basis for your answer. Answer: Yes
7	When did the activity occur?	Create a short timeline using exact PCAP timestamps. State the time zone displayed by Wireshark and any conversion you make. 94410 15759.950751 192.168.15.4 74.125.19.99 TCP 70 [TCP Retransmission] 36858 → 80 [FIN, ACK] Seq=514 Ack=594 Win=524280 Len=0 TSval=936588932 TSecr=1001670443 

No	Investigation question	Minimum evidence to discuss
8	What conclusion can you defend?	Based on the Wireshark evidence shown above the Packet 531 is a TCP retransmission from 69.17.116.124 to the internal host 192.168.1.64. This indicates that a TCP segment was retransmitted, which can occur when the original segment or its acknowledgement



Wireshark starting points

The following filters support the workflow shown in the case material. Replace the values in angle brackets with values you derive from the capture. Do not treat a filter result as a conclusion until you inspect the packet details and document the connection to the investigation question.

Purpose

Example display filter

Purpose	Example display filter
Locate web requests	<code>http.request.method == "GET" && http.host == "www.willselfdestruct.com"</code>
Search page content	<code>data-text-lines contains "secure anonymous E-mail"</code>
Identify form activity	<code>http.request.method == "POST"</code>
Narrow to the relevant exchange	<code>ip.src == <client_ip> && ip.dst == <service_ip> && http.request.method == "POST"</code>
Trace a device	<code>eth.addr == <client_mac></code>
Look for identity evidence	<code>eth.src == <client_mac> && http.cookie contains "@"</code>

Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and the case title.
2. Executive conclusion. State the supported conclusion in a short, cautious paragraph. Do not name a suspect without explaining the evidence link.
3. Evidence acquisition and integrity. Record the source, file name, size, hash, working copy, and any published checksum comparison.
4. Method. Explain the sequence of Wireshark filters, packet inspection, and stream review you used.
5. Findings. Answer each of the eight investigation questions with citations to packet numbers, timestamps, screenshots, and evidence-log IDs.
6. Timeline. Show the relevant sequence of requests and responses using the timestamps from the PCAP.
7. Attribution assessment. Distinguish observed facts from inferences, give a confidence level, and explain the shared-network limitation and any other alternative explanation.
8. Appendices. Include labelled screenshots and the completed evidence log. Screenshots must show the active filter, packet list, relevant packet detail, and time where possible.

Evidence log template

Include a completed version of this table in your report or evidence appendix. You may add rows where needed.

Evidence ID	Packet or item	Finding	Why it matters	Screenshot or appendix
E01	First packet: 2008-07-21 21:51:07			
E02	Last packet: 2008-07-22 02:13:47			

Evidence ID	Packet or item	Finding	Why it matters	Screenshot or appendix
E03	Packets 94410 94410 (100.0%)			
E04				
E05				
E06				

Submission rules

- Work individually. Your interpretation, report wording, screenshots, and evidence log must be your own.
- Submit the report as SBTDF204_CaseStudy1_RegistrationNumber.pdf or .docx. If you submit separate screenshots, use SBTDF204_CaseStudy1_Evidence_RegistrationNumber.zip.
- Do not upload the original PCAP unless your lecturer specifically requests it. Submit only the report and the supporting evidence required for assessment.
- Treat the packet capture as controlled training material. Use it only for this authorised academic case study. Do not contact, identify, or make allegations about real people outside the scenario.
- Redact unrelated personal data that may appear in screenshots. Keep enough packet detail visible for the examiner to verify your finding.

Marking guide

The E-Campus maximum score applies to this case study. The assessor will apply the following percentage weighting to that maximum score.

Criterion	What strong work demonstrates	Weight
Evidence acquisition and integrity	Records source, working copy, hash, and evidence handling clearly.	10%
Traffic discovery and filter use	Uses defensible filters and cites packets that identify the web service and client activity.	20%
Message correlation	Connects the HTTP evidence to the harassment content without overstating what the capture proves.	20%
Device and identity analysis	Uses MAC and cookie or browser evidence accurately and explains the link to the roster.	20%
Timeline and attribution assessment	Provides a chronological account, conclusion, confidence level, and relevant limitations.	20%
Report quality and evidence presentation	Writes clearly, labels screenshots, cites packet numbers, and submits a usable evidence log.	10%

Final check before submission

- Every major conclusion has a packet number, timestamp, filter, and screenshot or evidence-log reference.
- Your report explains why the client IP alone cannot establish the sender identity in a shared open wireless environment.

- Your timeline uses the time zone shown in Wireshark and identifies any conversion you made.
- Your conclusion stays within the scope of the evidence and clearly states the remaining uncertainty.

026-10-05 09:06:52 (312 KB/s) - 'nitroba.pcap' saved [56180821/56180821]

- MD5 9981827f11968773ff815e39f5458ec8
- SHA1 65656392412add15f93f8585197a8998aaeb50a1
- SHA256 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
- 2026-10-05 09:06:52 (312 KB/s) - 'nitroba.pcap' saved [56180821/56180821]